Aaryaveer Yadav

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SUMMARY

Motivated and detail-oriented Computer Science (AI & ML) undergraduate with expertise in Artificial Intelligence, Machine Learning, Data Analytics, and Full-Stack Web Development Strong foundation in Data Structures & Algorithms, Object-Oriented Programming, DBMS, and Software Engineering

EDUCATION

Vellore Institute of Technology, Bhopal University
Bachelor of Technology in Computer Science – AI & Machine Learning | Oct 2023 – Present

WORK EXPERIENCE

Intern – Professional Certification Program in AI and Data Science
DRISHTI CPS, IIT Indore, India
• Currently undergoing intensive training and hands-on projects in Artificial Intelligence and Data Science
• Gaining exposure to real-world applications of Machine Learning, Deep Learning, and Data Analytics
• Collaborating with research mentors and peers to solve domain-specific problems using Python and ML frameworks

UNIVERSITY PROJECTS

Face and Motion Detection System (LINK)

Python, TensorFlow, OpenCV

- **Designed a real-time system to detect human faces and motion in video streams**
- **Employed Haar cascades and background subtraction for efficient detection**
- **Optimized performance for live camera feed processing using OpenCV and TensorFlow**

ReRun – Autonomous AI Coding Agent (LINK)

Python, FastAPI, React, Docker, SQLite/PostgreSQL

- **Designed a hierarchical agent workflow to plan tasks, generate code, execute it, recover from runtime errors, and verify task completion.**
- **Used a Master Agent to coordinate specialized Planner, Coding, Recovery, and Task Validator agents.**
- **Planned sandboxed code execution with runtime monitoring and task-level verification to catch cases where code runs successfully but does not fulfill the request.**

Brain Tumor MRI Segmentation

Python, TensorFlow/Keras, U-Net, OpenCV

- **Developed a lightweight U-Net model to segment tumor regions from FLAIR brain MRI images.**
- **Prepared MRI images using normalization, resizing, filtering, CLAHE, and data augmentation.**
- **Evaluated segmentation using Dice score and IoU, reaching approximately 0.40 validation Dice and 0.32 IoU, compared with approximately 0.066 Dice for a classical image-processing baseline.**

TECHNICAL SKILLS

Programming Languages: Python, C++, JavaScript, SQL
Web Technologies: HTML5, CSS3, React.js, Flask, REST APIs
AI & Machine Learning: Machine Learning, Deep Learning, Computer Vision, TensorFlow, OpenCV, CNN, LSTM, Generative AI
Databases: MYSQL
Cloud & Tools: AWS Elastic Beanstalk, Git, GitHub, Power BI
Core CS Concepts: Data Structures & Algorithms, Object-Oriented Programming, DBMS, Computer Networks

CERTIFICATIONS

. Python Essentials
. Fundamentals of AI and ML
. Applied Machine Learning in Python
. Introduction to Internet of Things – Top 5% Elite
. Cloud Computing
. Computer Vision
Professional Certification Program in Al and Data Science from DRISHTI CPS IIT Indore